

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	saturday

SET 8 – EMPLOYEE PERFORMANCE ANALYSIS

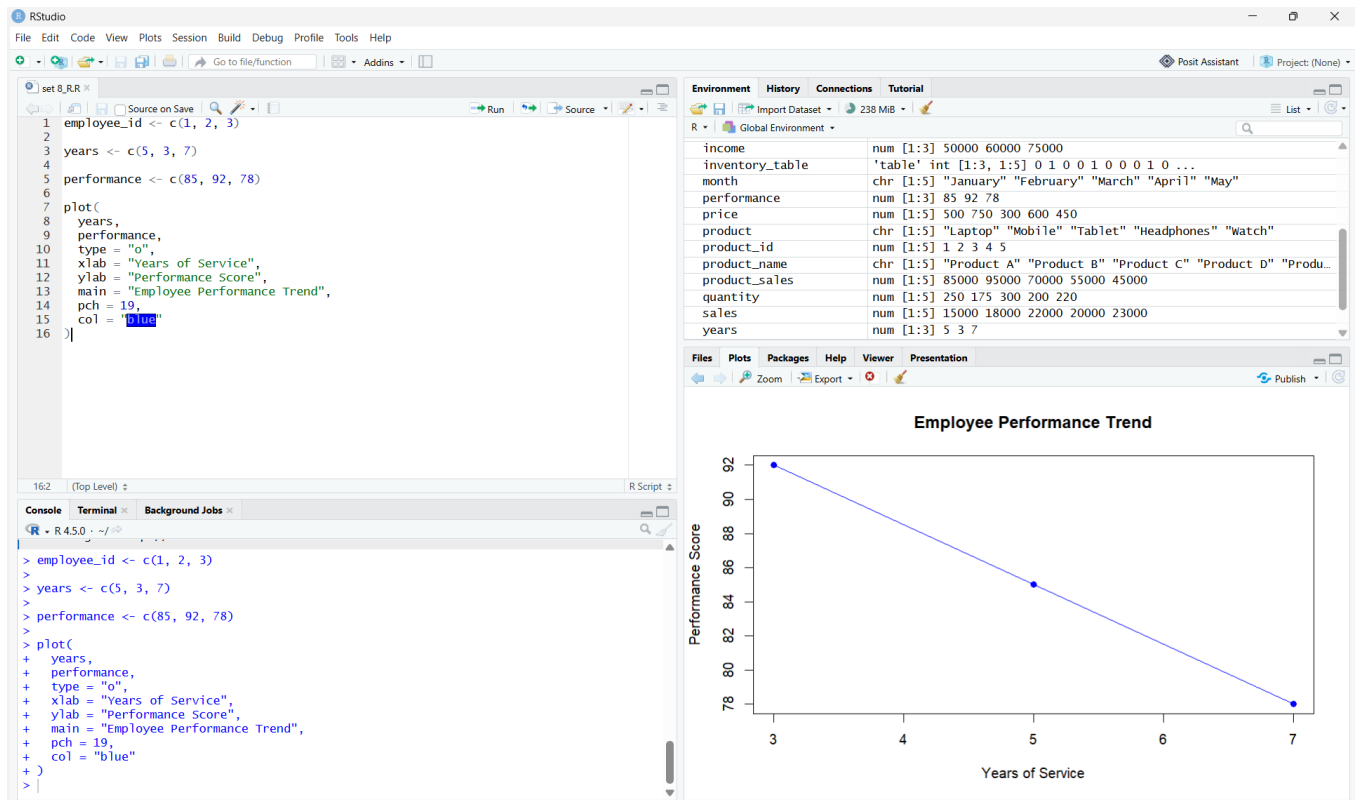
Aim

To analyze and visualize employee performance data using line charts, bar charts, tables, and an interactive dashboard.

PROGRAM 1: Line Chart – Employee Performance

To create a line chart to visualize the performance scores of employees based on their years of service.

```
employee_id <- c(1, 2, 3)
years <- c(5, 3, 7)
performance <- c(85, 92, 78)
plot(
  years,
  performance,
  type = "o",
  xlab = "Years of Service",
  ylab = "Performance Score",
  main = "Employee Performance Trend",
  pch = 19,
  col = "blue"
)
```



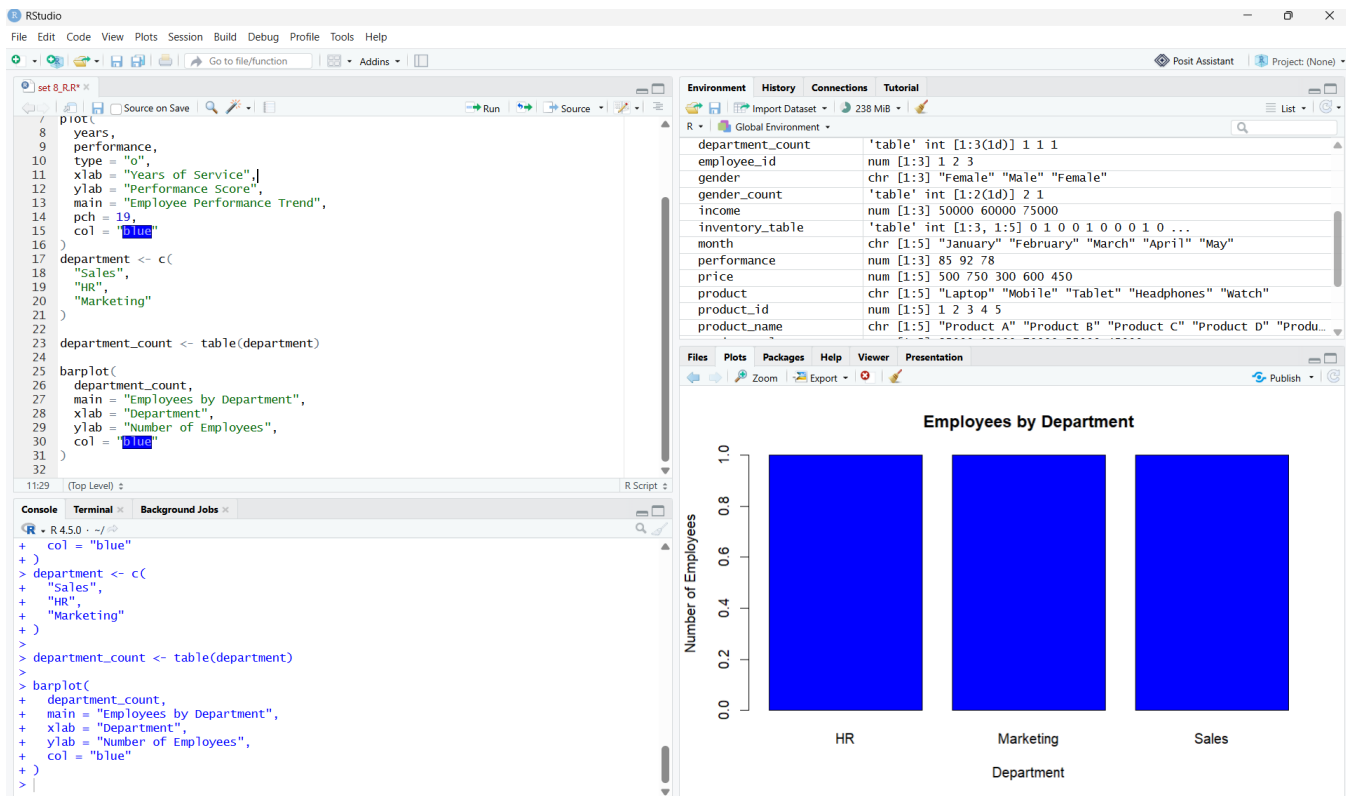
Output 1: Daily website page views

Observation: The highest performance score is **92** for the employee with 3 years of service. The employee with 7 years of service has a score of 78.

PROGRAM 2: Bar Chart – Employees by Department

To create a bar chart showing the distribution of employees across different departments.

```
department <- c(
  "Sales",
  "HR",
  "Marketing"
)
department_count <- table(department)
barplot(
  department_count,
  main = "Employees by Department",
  xlab = "Department",
  ylab = "Number of Employees",
  col = "blue"
)
```



Output 2: click through rate by day

Observation: Each department contains **one employee** in the given dataset.

PROGRAM 3: Table – Employee Performance Data

To create a table displaying the performance information of each employee.

```

employee_id <- c(1, 2, 3)
department <- c(
  "Sales",
  "HR",
  "Marketing"
)
years <- c(5, 3, 7)
performance <- c(85, 92, 78)
employee_table <- data.frame(
  Employee_ID = employee_id,
  Department = department,
  Years_of_Service = years,
  Performance_Score = performance
)
employee_table

```

The screenshot displays the RStudio interface. The top pane shows an R script with the following code:

```
28 xlab = "department",
29 ylab = "Number of Employees",
30 col = "blue"
31 )
32
33 employee_id <- c(1, 2, 3)
34
35 department <- c(
36   "Sales",
37   "HR",
38   "Marketing"
39 )
40
41 years <- c(5, 3, 7)
42
43 performance <- c(85, 92, 78)
44
45 employee_table <- data.frame(
46   Employee_ID = employee_id,
47   Department = department,
48   Years_of_Service = years,
49   Performance_Score = performance
50 )
51
52 employee_table
53
```

The bottom pane shows the console output, which includes the commands entered and the resulting data frame:

```
>
> years <- c(5, 3, 7)
>
> performance <- c(85, 92, 78)
>
> employee_table <- data.frame(
+   Employee_ID = employee_id,
+   Department = department,
+   Years_of_Service = years,
+   Performance_Score = performance
+ )
>
> employee_table
  Employee_ID Department Years_of_Service Performance_Score
1           1     Sales                5                 85
2           2        HR                 3                 92
3           3   Marketing                7                 78
> |
```

Output 3: user interactions

Observation: The table successfully displays the complete performance information of all three employees.

PROGRAM 4: Interactive Dashboard

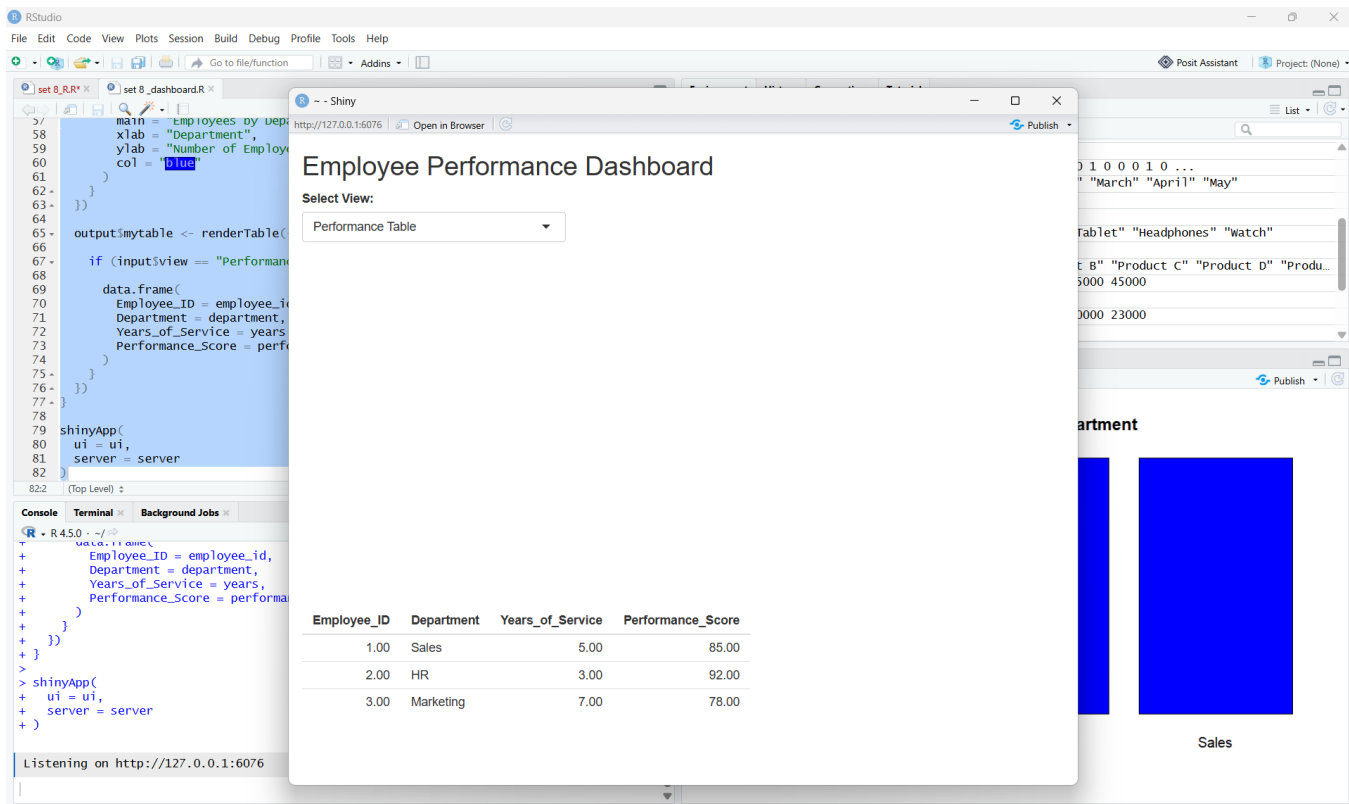
To develop an interactive dashboard combining the employee performance line chart, department bar chart, and performance table..

```
library(shiny)
employee_id <- c(1, 2, 3)
department <- c(
  "Sales",
  "HR",
  "Marketing"
)
years <- c(5, 3, 7)
performance <- c(85, 92, 78)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  selectInput(
    "view",
    "Select View:",
    choices = c(
      "Performance Trend",
      "Department Distribution",
      "Performance Table"
    )
  ),
  plotOutput("myplot"),
  tableOutput("mytable")
)
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (input$view == "Performance Trend") {
      plot(
        years,
        performance,
        type = "o",
        xlab = "Years of Service",
        ylab = "Performance Score",
        main = "Employee Performance Trend",
```

```

      pch = 19,
      col = "blue"
    )
  }
  if (input$view == "Department Distribution") {
    department_count <- table(department)
    barplot(
      department_count,
      main = "Employees by Department",
      xlab = "Department",
      ylab = "Number of Employees",
      col = "blue"
    )
  }
})
output$mytable <- renderTable({
  if (input$view == "Performance Table") {
    data.frame(
      Employee_ID = employee_id,
      Department = department,
      Years_of_Service = years,
      Performance_Score = performance
    )
  }
})
}
shinyApp(
  ui = ui,
  server = server
)

```



Output 4: website analytics dashboard

Observation: The dashboard allows the user to switch between employee performance trends, department distribution, and employee performance information.

Set 8 Procedure

- Create vectors for employee ID, department, years of service, and performance score.
- Use `plot()` to visualize employee performance based on years of service.
- Use `table()` and `barplot()` to display employees by department.
- Use `data.frame()` to create the employee performance table.
- Use Shiny to combine the line chart, bar chart, and table into an interactive dashboard.
- Save each question as a separate R script and run the programs in RStudio.

Set 8 Result

Employee performance data was successfully analyzed using a line chart, bar chart, and table. An interactive dashboard was also developed to explore employee performance and department information.